AI 3000/7010: Reinforcement Learning
Tutorial 6

For all questions, we work with an infinite-horizon MDP with discount factor $\lambda < 1$, finite state and action spaces $\mathcal{S}$ and $\mathcal{A}$ respectively, and time homogeneous transition matrix $P(s' \mid s, a)$.

1 Second Moment

(Adapted from Mannor, Mansour & Tamar)

Consider a stationary policy $(\phi, \phi, \ldots)$. Recall that the value function for this policy is

$$v^{\phi}_{\infty,\lambda}(s) = \mathbb{E}^{\phi}\left[\sum_{n=1}^{\infty} \lambda^{n-1} R_n \,\middle|\, S_0 = s\right]$$

and that we can rewrite this as a vector equation

$$v^{\phi}_{\infty,\lambda} = r_{\phi} + \lambda P_{\phi} v^{\phi}_{\infty,\lambda} \qquad \Longleftrightarrow \qquad v^{\phi}_{\infty,\lambda} = L_{\phi} v^{\phi}_{\infty,\lambda}$$

Now consider the quantity $m^{\phi}_{\infty,\lambda}$ defined similarly as

$$m^{\phi}_{\infty,\lambda}(s) := \mathbb{E}^{\phi}\left[\left(\sum_{n=1}^{\infty} \lambda^{n-1} R_n\right)^2 \,\middle|\, S_0 = s\right]$$

1. Write a similar vector equation for $m^{\phi}_{\infty,\lambda}$.
2. Keeping $v^{\phi}_{\infty,\lambda}$ fixed, express $m^{\phi}_{\infty,\lambda}$ as the fixed point of an operator T_{ϕ}. Is T_{ϕ} a contraction?

2 History Dependent Optimal Policies

(Adapted from Puterman & Chan, 2026)

The original question asks you to go from Π^{HR} to Π^{SR} in a similar manner. This extension is left as an exercise.

Let $(\phi_0, \phi_1, \phi_2, \ldots) \in \Pi^{\mathrm{HD}}$ be optimal. Show that $(\phi_0, \phi_0, \phi_0, \ldots) \in \Pi^{\mathrm{SD}}$ is also optimal.

3 Interpreting Bellman Residual Minimization

Let $\{v_\theta\}_{\theta \in \Theta}$ be a parametric family of vectors in $\mathbb{R}^{|\mathcal{S}|}$. A student attempts to justify Bellman Residual Minimization to approximate $v^{\phi}_{\infty,\lambda}$ (for a fixed $\phi \colon \mathcal{S} \to \mathcal{A}$) as follows

$$\begin{aligned}
\arg\min_{\theta} \|v_\theta - L_\phi v_\theta\| &= \arg\min_{\theta} \|v_\theta - r_\phi - \lambda P_\phi v_\theta\| && (1)\\
&= \arg\min_{\theta} \|(I - \lambda P_\phi) v_\theta - r_\phi\| && (2)\\
&= \arg\min_{\theta} \|(I - \lambda P_\phi)\,(v_\theta - v^{\phi}_{\infty,\lambda})\| && (3)\\
&= \arg\min_{\theta} \|(I - \lambda P_\phi)\|\,\|v_\theta - v^{\phi}_{\infty,\lambda}\| && (4)\\
&= \arg\min_{\theta} \|v_\theta - v^{\phi}_{\infty,\lambda}\| && (5)
\end{aligned}$$

Is this interpretation correct? If not, where is the mistake and how can you correct it? Can you extend this justification for L instead of L_ϕ?